第七章 玻尔兹曼统计

§7.1 热力学量的统计表达式

1. 内能: $U = \sum a_l \varepsilon_l = \sum \omega_l \varepsilon_l e^{-\alpha-\beta\varepsilon_l}$

引入配分函数: $Z_1 := \sum \omega_l e^{-\beta\varepsilon_l}$ 1: 单粒子

⇒ 总粒子数 $N = \sum a_l = e^{-\alpha} Z_1$

$U = e^{-\alpha}\sum \varepsilon_l \omega_l e^{-\beta\varepsilon_l} = -e^{-\alpha}\frac{\partial}{\partial\beta}(Z_1) = -\frac{N}{Z_1}\frac{\partial Z_1}{\partial\beta}$

$= -N\frac{\partial \ln Z_1}{\partial\beta}$

2. 广义力 Y

$dU = dW + đQ = Y\,dy + đQ$

$Y = \sum \frac{\partial\varepsilon_l}{\partial y}\cdot a_l = \overline{\frac{\partial\varepsilon_l}{\partial y}}\cdot N$

$= \sum \frac{\partial\varepsilon_l}{\partial y}\cdot\omega_l e^{-\alpha-\beta\varepsilon_l} = e^{-\alpha}\sum\frac{\partial\varepsilon_l}{\partial y}\cdot\frac{\partial Z_1}{\partial\varepsilon_l}\cdot\left(-\frac{1}{\beta}\right)$

$= -\frac{e^{-\alpha}}{\beta}\frac{\partial Z_1}{\partial y} = -\frac{N}{\beta Z_1}\frac{\partial Z_1}{\partial y}$

$= -\frac{N}{\beta}\frac{\partial(\ln Z_1)}{\partial y}$

对于 Y: p, $y = -V$, $p = \frac{N}{\beta}\frac{\partial(\ln Z_1)}{\partial V}$

$Y\,dy = dy\cdot\sum\frac{\partial\varepsilon_l}{\partial y}\cdot a_l = \sum a_l\,d\varepsilon_l$

$dU = d(\sum a_l\varepsilon_l) = \sum a_l\,d\varepsilon_l + \sum\varepsilon_l\,da_l = Y\,dy + đQ$

⇒ $đQ = \sum\varepsilon_l\,da_l$ 热量: 由于粒子重新分布引起的能量变化.

3. 熵 S

$dS = \frac{đQ}{T} = \frac{1}{T}(dU - dW)$

$= -\frac{N}{T}\left[d\left(\frac{\partial(\ln Z_1)}{\partial\beta}\right) - \frac{1}{\beta}\frac{\partial(\ln Z_1)}{\partial y}dy\right]$

⇒ $\beta(dU - dW) = -N\beta\,d\left(\frac{\partial(\ln Z_1)}{\partial\beta}\right) + N\frac{\partial\ln Z_1}{\partial y}dy$ → β 为积分因子

$Z_1 = Z_1(\beta, y)$ 为什么 Z_1 是 y 的函数? A: 对简单系统: $\beta\sim T$, $y\sim V$

$d(\ln Z_1) = \frac{\partial(\ln Z_1)}{\partial y}dy + \frac{\partial(\ln Z_1)}{\partial\beta}d\beta$

$= N[d\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta} d\beta - d(\frac{\partial \ln Z_1}{\partial \beta})\beta] = N d(\ln Z_1 - \beta \cdot \frac{\partial \ln Z_1}{\partial \beta})$

一个函数的任两个积分因子之比为 S 的函数

令 $\beta = \frac{1}{kT}$

β仅与T有关 ⇒ k为常量.

$dS = Nk\, d(\ln Z_1 - \beta \frac{\partial \ln Z_1}{\partial \beta})$

$\Rightarrow S = Nk(\ln Z_1 - \beta \frac{\partial \ln Z_1}{\partial \beta})$

对于符合玻尔兹曼分布的系统

$Z_1 = Ne^{\alpha}$ $\quad \ln Z_1 = \ln N + \alpha$

$= Nk(\ln N + \alpha - \beta \frac{\partial \ln Z_1}{\partial \beta})$

$= k(N\ln N + \alpha N + \beta U)$

$= k[N\ln N + \sum_l(\alpha + \beta\varepsilon_l)a_l]$ $\quad a_l = \omega_l e^{-\alpha-\beta\varepsilon_l}$

$= k[N\ln N + \sum_l \ln(\frac{\omega_l}{a_l}) a_l]$

$= k\ln \Omega_{M.B.}$

$= k\ln(\frac{\Omega_{M.B.}}{N!})$ 经典极限下的玻色/费米系统.

以 T、V 为变量:

特性函数: $F = U - TS$

$= -N\frac{\partial \ln Z_1}{\partial \beta} - NkT(\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta}\cdot\beta)$

$= -\frac{N}{\beta}\ln Z_1$

$= -\frac{N}{\beta}\ln Z_1 + \frac{1}{\beta}\ln N!$ 经典极限下的玻色/费米系统.

$\Omega = N\ln N - a_l\ln(\frac{\omega_l}{a_l})$, $Z = e^{\alpha}\cdot N$

$\to \Omega \to \frac{\Omega}{N!}$, $Z_1^N \to \frac{Z_1^N}{N!}$?

配分函数 Z ⇒ U、Y、S、F 的统计表达式

经典统计:

$Z_1 = \sum_l \frac{\Delta\omega_l}{h_0^r} e^{-\beta\varepsilon_l}$

$= \int e^{-\beta\varepsilon_l} \frac{d\omega}{h_0^r}$

$a_l = e^{-\alpha-\beta\varepsilon_l}\frac{\Delta\omega_l}{h_0^r} = \frac{N}{Z_1} e^{-\beta\varepsilon_l}\frac{\Delta\omega_l}{h_0^r}$

> U、Y 与 h_0 无关 (都是 $\frac{\partial}{\partial}(\ln Z_1)$)

> S 与 h 有关 (含有 $\ln Z_1$)

§ 7.2 理想气体的物态方程

对理想气体：

$$z_1 = \int e^{-\frac{\beta}{2m}(p_x^2+p_y^2+p_z^2)} \frac{dp_x dp_y dp_z dx dy dz}{h^3}$$

$$= \frac{1}{h^3}\int dx dy dz \int_{-\infty}^{+\infty} e^{-\frac{\beta}{2m}p_x^2} dp_x \cdot \int_{-\infty}^{+\infty} e^{-\frac{\beta}{2m}p_y^2} dp_y \cdot \int_{-\infty}^{+\infty} e^{-\frac{\beta}{2m}p_z^2} dp_z$$

$$\int_0^{+\infty} e^{-\frac{\beta}{2m}p^2} dp = \sqrt{\frac{2m}{\beta}} \cdot \frac{\sqrt{\pi}}{2}$$

$$= V \cdot \left(\frac{2\pi m}{h^2\beta}\right)^{3/2}$$

$$\hookrightarrow \quad p = \frac{N}{\beta} \cdot \frac{\partial \ln z_1}{\partial V} = \frac{N}{\beta} \cdot \frac{1}{V} = \frac{NkT}{V} = \frac{nRT}{V}$$

经典极限条件：$e^{\alpha} = \frac{z_1}{N} = \frac{V}{N}\left(\frac{2\pi m}{h^2\beta}\right)^{3/2} \gg 1$

$\varepsilon \sim kT$, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m\varepsilon}} \sim \frac{1}{\sqrt{kT}} \sim \sqrt{\beta}$

$\Rightarrow \quad n\lambda^3 \ll 1$

§7.3 麦克斯韦速度分布律

$$a_l = \omega_l e^{-\alpha-\beta\varepsilon_l} = \frac{dx\,dy\,dz\,dp_x\,dp_y\,dp_z}{h^3} e^{-\alpha-\beta\varepsilon_l}$$

$$\varepsilon = \frac{1}{2m}(p_x^2+p_y^2+p_z^2)$$

在 $p_x \sim p_x+dp_x$，$p_y \sim p_y+dp_y$，$p_z \sim p_z+dp_z$ 内的粒子数

$\int dx\,dy\,dz \to$ $$\frac{V}{h^3} e^{-\alpha-\frac{\beta}{2m}(p_x^2+p_y^2+p_z^2)}\,dp_x\,dp_y\,dp_z$$

归一化条件：$N = \sum a_l$

$$= \frac{V}{h^3}e^{-\alpha}\iiint e^{-\frac{\beta}{2m}(p_x^2+p_y^2+p_z^2)}\,dp_x\,dp_y\,dp_z$$

$$= \frac{Ve^{-\alpha}}{h^3}\int_0^{+\infty} 4\pi p^2 e^{-\frac{\beta p^2}{2m}}\,dp$$

$$= \frac{4\pi V e^{-\alpha}}{h^3}\cdot\frac{\sqrt{\pi}}{4\left(\frac{\beta}{2m}\right)^{3/2}} = \left(\frac{2m\pi kT}{h^2}\right)^{3/2} V e^{-\alpha}$$

$$\Rightarrow e^{\alpha} = \left(\frac{2m\pi kT}{h^2}\right)^{3/2}\frac{V}{N}$$

在 $p_x \sim p_x+dp_x$，$p_y \sim p_y+dp_y$，$p_z \sim p_z+dp_z$ 内的粒子数

$$dN = N\left(\frac{1}{2\pi mkT}\right)^{3/2} e^{-\frac{1}{2mkT}(p_x^2+p_y^2+p_z^2)}\,dp_x\,dp_y\,dp_z$$

$$= N\left(\frac{m}{2\pi kT}\right)^{3/2} e^{-\frac{m}{2kT}(v_x^2+v_y^2+v_z^2)}\,dv_x\,dv_y\,dv_z$$

$\Rightarrow$ 速度的概率密度函数 $f(v_x, v_y, v_z)$

$$f(v_x, v_y, v_z)\,dv_x\,dv_y\,dv_z = \frac{dN}{V}$$

$$= n\left(\frac{m}{2\pi kT}\right)^{3/2} e^{-\frac{m}{2kT}(v_x^2+v_y^2+v_z^2)}\,dv_x\,dv_y\,dv_z$$

归一化条件 $\iiint f(v_x, v_y, v_z)\,dv_x\,dv_y\,dv_z$

球坐标：$f(v) = 4\pi n\cdot\left(\frac{m}{2\pi kT}\right)^{3/2} e^{-\frac{mv^2}{2kT}}\, v$

§ 7.4 能量均分定理

对于处在温度 T 的平衡状态的经典系统，粒子能量中每一个独立的平方项的平均值为 $\frac{1}{2}kT$

证明：粒子动能 $\varepsilon_p = \frac{1}{2}\sum_i a_i p_i^2$

$$\overline{\frac{1}{2}a_i p_i^2} = \frac{a_i}{2N}\int p_i^2 \underline{e^{-\alpha-\beta\cdot\frac{1}{2}a_i p_i^2}\,\frac{dq_1\cdots dq_r\,dp_1\cdots dp_r}{h_0^r}} \quad a_i\ (经典统计)$$

$Z_1 = \frac{N}{e^{-\alpha}}$

$$= \frac{a_i}{2Z_1}\int_{-\infty}^{+\infty} p_i^2 e^{-\frac{\beta}{2}a_i p_i^2}\,dp_i \cdot \int \frac{dq_1\cdots dq_r\,dp_1\cdots dp_r}{h_0^r\cdot dp_i}$$

$$= \frac{-1}{2\beta Z_1}\int_{-\infty}^{+\infty} p_i\, d\left(e^{-\frac{1}{2}\beta a_i p_i^2}\right)\int \frac{dq_1\cdots dq_r\,dp_1\cdots dp_r}{h_0^r\cdot dp_i}$$

$$= \frac{1}{2\beta Z_1}\left(p_i e^{-\frac{1}{2}\beta a_i p_i^2}\Big|_{+\infty}^{-\infty} + \int_{-\infty}^{+\infty} e^{-\frac{1}{2}\beta a_i p_i^2}\,dp_i \right)\int \frac{dq_1\cdots dq_r\,dp_1\cdots dp_r}{h_0^r\cdot dp_i}$$

$$= \frac{1}{2\beta Z_1}\int e^{-\beta\varepsilon}\,\frac{dq_1\cdots dq_r\,dp_1\cdots dp_r}{h_0^r}$$

$Z_1 = \sum \omega_l e^{-\beta\varepsilon} = \sum \frac{\Delta\omega}{h_0^r} e^{-\beta\varepsilon}$

$$= \frac{1}{2\beta} = \frac{1}{2}kT$$

对于粒子势能，$\varepsilon_q = \frac{1}{2}\sum_i b_i q_i^2 + \varepsilon_q(q_i')$

亦有 $\overline{\frac{1}{2}b_i q_i^2} = \frac{1}{2}kT$

例：

(1) 单原子分子：

$\varepsilon = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2)$

$\to \bar{\varepsilon} = \frac{3}{2}kT$，$U = N\bar{\varepsilon} = \frac{3}{2}NkT$

$C_V = \frac{dU}{dT} = \frac{3}{2}Nk$，$C_p = C_V + Nk = \frac{5}{2}Nk$，$\gamma = \frac{C_p}{C_V} = \frac{5}{3}$

(2) 刚性双原子分子：

$$\varepsilon = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + \frac{1}{2I}\left(p_\theta^2 + \frac{1}{\sin^2\theta}p_\varphi^2\right) + \underbrace{\frac{p_r^2}{2\mu} + u(r)}_{=0\ (刚性)}$$

$\bar{\varepsilon} = \frac{5}{2}kT$，$U = \frac{5}{2}NkT$，$C_V = \frac{5}{2}Nk$，$C_p = \frac{7}{2}Nk$，$\gamma = \frac{7}{5}$

(3) 固体

$\varepsilon = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + \frac{1}{2}m\omega^2(q_x^2 + q_y^2 + q_z^2)$

$\bar{\varepsilon} = 3kT$，$U = 3NkT$

$C_V = 3Nk$　　* 自由电子不提供热容？

平衡辐射问题：

$$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{r} - \omega t)}$$

周期性边界条件：　$L = n\lambda$

$$k_x = \frac{2\pi}{L} n_x, \quad k_y = \frac{2\pi}{L} n_y, \quad k_z = \frac{2\pi}{L} n_z$$

$$\omega = ck$$

1个 k → 1个振动自由度

在体积 V，$k_x \sim k_x + dk_x$，$k_y \sim k_y + dk_y$，$k_z \sim k_z + dk_z$ 的振动自由度数.

$$dn_x\, dn_y\, dn_z = 2\cdot\left(\frac{L}{2\pi}\right)^3 dk_x\, dk_y\, dk_z$$　　2个偏振方向

$$= \frac{L^3}{4\pi^3 c^3}\, d\omega_x\, d\omega_y\, d\omega_z$$

谱密度

$$:= D(\omega)\,d\omega = \frac{L^3}{\pi^2 c^3}\,\omega^2\, d\omega$$

对振动自由度：　$\bar{\varepsilon} = kT$

内能密度　$u(\omega)\,d\omega$

$$= D(\omega)\,d\omega \cdot kT = \frac{L^3 kT}{\pi^2 c^3}\,\omega^2\, d\omega$$　瑞利—金斯公式.

总　$u = \int u(\omega)\, d\omega$

$$= \frac{L^3 kT}{\pi^2 c^3}\int_0^{+\infty} \omega^2\, d\omega \to \infty$$

与 $u = aT^4 \cdot V$ 矛盾.

§ 7.5 理想气体的内能和热容

1. 量子理论

能量 $\varepsilon = \varepsilon^t + \varepsilon^v + \varepsilon^r$

配分函数 $Z_1 = \sum_l \omega_l e^{-\beta\varepsilon_l}$

$$= \sum_{t,v,r} \omega^t \cdot \omega^v \cdot \omega^r \, e^{-\beta(\varepsilon^t+\varepsilon^v+\varepsilon^r)}$$

$$= Z_1^t \cdot Z_1^v \cdot Z_1^r$$

$$\Rightarrow U = -N \cdot \frac{\partial}{\partial\beta}(\ln Z_1) = -N \cdot \frac{\partial}{\partial\beta}(\ln Z_1^t + \ln Z_1^v + \ln Z_1^r)$$

$$= U^t + U^v + U^r$$

$$C_V = C_V^t + C_V^v + C_V^r$$

(1) 对于平动：

$$Z_1^t = \int \frac{dx\,dy\,dz \cdot dp_x\,dp_y\,dp_z}{h^3} \cdot e^{-\beta\frac{1}{2m}(p_x^2+p_y^2+p_z^2)}$$

$$= \frac{V}{h^3} \cdot \left(2 \cdot \frac{\sqrt{\pi}}{2\sqrt{\frac{\beta}{2m}}}\right)^3 = \left(\frac{2\pi m}{h^2\beta}\right)^{3/2} V$$

$$U^t = \frac{3}{2}N \cdot \frac{1}{\beta} = \frac{3}{2}NkT$$

$$C_V^t = \frac{3}{2}Nk$$

(2) 对于振动：

认为两原子的相对振动可视为线性谐振子

$$\varepsilon_n = (n+\tfrac{1}{2})\hbar\omega$$

$$Z_1^v = \sum_{n=0}^{\infty} 1 \cdot e^{-\beta(n+\frac{1}{2})\hbar\omega}$$

每个能级上只有一个谐振子？

$$= \frac{e^{-\frac{1}{2}\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}}$$

$$U^v = -N \cdot \frac{\partial}{\partial\beta} \ln Z_1^v = N\left(\frac{1}{2}\hbar\omega + \frac{-e^{-\beta\hbar\omega}\hbar\omega}{1-e^{-\beta\hbar\omega}}\right)$$

$$= N\hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega}-1}\right)$$ 零点能量 + 热激发能量.

$$C_V^v = \left(\frac{\partial U}{\partial T}\right)_V = \frac{\partial U}{\partial\beta} \cdot \frac{\partial\beta}{\partial T} = -\frac{1}{kT^2} \cdot N\hbar\omega \, \frac{-\hbar\omega \cdot e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2}$$

$$= \frac{N\beta(\hbar\omega)e^{\cdots}}{(e^{\beta\hbar\omega}-1)^2} = Nk\left(\frac{\hbar\omega}{kT}\right)^2 \frac{e}{(e^{\hbar\omega/kT}-1)^2}$$

特征温度： $k\theta_v = \hbar\omega$ $\theta_v \approx 10^3 k$, $T \ll \theta_v$

$$u^v = Nk\theta_v \cdot \left(\frac{1}{2} + \frac{1}{e^{\frac{\theta_v}{T}}-1}\right) \approx Nk\theta_v\left(\frac{1}{2} + e^{-\frac{\theta_v}{T}}\right)$$

$$C_V^v = Nk\left(\frac{\theta_v}{T}\right)^2 \frac{e^{\frac{\theta_v}{T}}}{(e^{\frac{\theta_v}{T}}-1)^2}$$

$$\approx Nk\,\frac{\theta_v}{T}\cdot e^{-\frac{\theta_v}{T}} \approx 0$$

解释：常温下 $\hbar\omega \gg kT$，振子跃迁概率极小
都处于基态，即"振动自由度被冻结"

(3) 对于转动：

$$\varepsilon^r = \frac{l(l+1)\hbar^2}{2I}$$

$$Z_1^r = \sum_{l=0}^{\infty}(2l+1)e^{-\beta\frac{l(l+1)\hbar^2}{2I}}$$ $k\theta_r = \frac{\hbar^2}{2I}$

$$= \sum_{l=0}^{\infty}(2l+1)e^{-(l+1)l\left(\frac{\theta_r}{T}\right)}$$ $\theta_r \ll T$ $x = (l+1)l\frac{\theta_r}{T}$

↳ l 改变时，x 近似连续

$$\approx \frac{T}{\theta_r}\int_0^{+\infty} e^{-x}\,dx = \frac{T}{\theta_r} = \frac{2I}{\beta\hbar^2}$$

$$u^r = -N\cdot\frac{\partial}{\partial\beta}(\ln Z_1^r)$$

$$= N\frac{1}{\beta} = NkT$$

$$C_V^r = \frac{\partial u^r}{\partial T} = Nk$$

对同核双分子 $[H_2, N_2 \cdots]$

$$\varepsilon^r = \frac{(l+1)l\hbar^2}{2I}$$ $\begin{cases} l \text{ 为奇数 正氢 O 占}\frac{3}{4} \\ l \text{ 为偶数 仲氢 P 占}\frac{1}{4} \end{cases}$ why?

$$\begin{cases} Z_o^r = \sum\limits_{l\text{为奇}}(2l+1)e^{-\beta\frac{l(l+1)\theta_r}{T}} \\ Z_p^r = \sum\limits_{l\text{为偶}}(2l+1)e^{-\beta\frac{l(l+1)\theta_r}{T}} \end{cases}$$

$$u^r = -N\left[\frac{3}{4}\frac{\partial}{\partial\beta}(\ln Z_o^r) + \frac{1}{4}\frac{\partial}{\partial\beta}(\ln Z_p^r)\right]$$

$\theta_r(H_2) \approx 85.4k$

高温：$\sum_{l为奇} \approx \sum_{l为偶} = \frac{1}{2}\sum_{l=0}^{\infty}$ $\quad T \gg \theta_r \quad C_V^r = Nk$

低温：C_V^r 复杂

(4) 电子

2. 经典统计

$$\varepsilon = \varepsilon^t + \varepsilon^v + \varepsilon^r$$

$$= \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + \frac{1}{2I}\left(p_\theta^2 + \frac{1}{\sin^2\theta} p_\varphi^2\right) + \frac{1}{2\mu}(p_r^2 + \mu^2\omega^2 r^2)$$

$$Z = \int e^{-\beta\varepsilon} \frac{dq_1 \cdots dq_r \, dp_1 \cdots dp_r}{h_0^r}$$

$$= \int e^{-\beta\varepsilon^t} \cdot \frac{dq_1 \cdots dq_r \, dp_1 \cdots dp_r}{h_0^r} \int e^{-\beta\varepsilon^v} \cdot \frac{dq_1 \cdots dq_r \, dp_1 \cdots dp_r}{h_0^r}$$

$$\int e^{-\beta\varepsilon^r} \cdot \frac{dq_1 \cdots dq_r \, dp_1 \cdots dp_r}{h_0^r}$$

$$= Z^t \cdot Z^v \cdot Z^r$$

$$Z_1^t = \int e^{-\frac{\beta}{2m}(p_x^2 + p_y^2 + p_z^2)} \frac{dx\,dy\,dz\,dp_x\,dp_y\,dp_z}{h_0^3}$$

$$= \frac{4\pi V}{h_0^3}\int_0^{+\infty} p^2 e^{-\frac{\beta}{2m}p^2} dp = \frac{4\pi^2}{h_0^3} \cdot \frac{\sqrt{\pi}}{4\left(\frac{\beta}{2m}\right)^{3/2}}$$

$$= V \cdot \left(\frac{2m\pi}{h_0^2\beta}\right)^{3/2}$$

$$Z_1^v = \int e^{-\frac{\beta}{2\mu}(p_r^2 + \mu^2\omega^2 r^2)} \frac{dr\,dp_r}{h_0}$$

$$= \frac{1}{h_0}\int_{-\infty}^{+\infty} e^{-\frac{\beta}{2}\mu\omega^2 r^2} dr \cdot \int_{-\infty}^{+\infty} e^{-\frac{\beta}{2\mu}p_r^2} dp_r$$

$$= \frac{1}{h_0} \cdot \frac{2\sqrt{\pi}}{2\sqrt{\frac{1}{2}\mu\omega^2\beta}} \cdot \frac{2\sqrt{\pi}}{2\cdot\sqrt{\frac{1}{2\mu}\beta}}$$

$$= \frac{2\pi}{h_0\beta\omega}$$

$$Z_1^r = \int e^{-\frac{\beta}{2I}\left(p_\theta^2 + \frac{1}{\sin^2\theta}p_\varphi^2\right)} \cdot \frac{d\theta\,d\varphi\,dp_\theta\,dp_\varphi}{h_0^2}$$

$$= \frac{1}{h_0^2}\int_0^{\pi} d\theta \int_0^{2\pi} d\varphi \cdot \int_{-\infty}^{+\infty} e^{-\frac{\beta}{2I}p_\theta^2} dp_\theta \int_{-\infty}^{+\infty} e^{-\frac{\beta}{2I\sin^2\theta}p_\varphi^2} dp_\varphi$$

$$= \frac{2\pi}{h_0^2}\int_0^{\pi} \frac{\sqrt{\pi}}{2\sqrt{\frac{\beta}{2I}}} \cdot \frac{\sqrt{\pi}}{2\sqrt{\frac{\beta}{2I\sin^2\theta}}} d\theta$$

$$= \frac{4\pi^2 I}{h_0^2}\int_0^{\pi} \sin\theta\, d\theta = \frac{8\pi^2 I}{h_0^2\beta}$$

$$U = -N\frac{\partial}{\partial\beta}(\ln z^t + \ln z^v + \ln z^r) \qquad C_V = \frac{7}{2}Nk$$

$$= \left(\frac{3}{2}+1+1\right)\frac{N}{\beta} = \frac{7}{2}NkT$$

§7.6 理想气体的熵

对于单原子气体:

$$z = V\cdot\left(\frac{2\pi m}{h^2\beta}\right)^{3/2}$$

$$S = Nk\left(\ln z - \beta\frac{\partial \ln z}{\partial\beta}\right)$$

$$= Nk\left[\ln V + \frac{3}{2}\ln\left(\frac{2\pi m}{h^2\beta}\right) + \frac{3}{2}\right]$$

$$= \frac{3}{2}Nk\ln T + Nk\ln V + \frac{3}{2}Nk\left[1+\ln\left(\frac{2\pi mk}{h^2}\right)\right]$$

此时熵不是广延量。

修正：考虑微观粒子全同性，在经典极限下：

$$S = k\ln\left(\frac{\Omega_{M.B}}{N!}\right)$$

$$\Rightarrow S = Nk\left(\ln z - \beta\frac{\partial\ln z}{\partial\beta}\right) - k\ln N!$$

$$\approx Nk\left(\ln z - \beta\frac{\partial\ln z}{\partial\beta}\right) - kN(\ln N - 1)$$

$$= \frac{3}{2}Nk\ln T + Nk\ln V + \frac{3}{2}Nk\left[1+\ln\left(\frac{2\pi mk}{h^2}\right)\right] - kN(\ln N - 1)$$

$$= \frac{3}{2}Nk\ln T + Nk\ln\left(\frac{V}{N}\right) + \frac{3}{2}Nk\left[\frac{5}{3} + \ln\frac{2\pi mk}{h^2}\right]$$

蒸气压:

化学势:

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}$$

$$F = -\frac{N}{\beta}\ln z + \frac{1}{\beta}\ln N! \approx -\frac{N}{\beta}[\ln z - (\ln N - 1)]$$

$\mu = -\frac{1}{\beta} \cdots$

$= -kT \ln\left(\frac{V\left(\frac{2\pi m}{h^2\beta}\right)^{3/2}}{N}\right) = kT \ln\left[\frac{N}{V}\left(\frac{h^2}{2\pi m kT}\right)^{3/2}\right]$

$= -kT \ln e^{\alpha} < 0$　　经典极限条件：$e^{\alpha} = \frac{Z_1}{N} \gg 1$

理想气体化学势为负

§7.7 固体热容的爱因斯坦理论

固体中原子的热运动可视为 $3N$ 个振子的振动

$\varepsilon = \hbar\omega(n+\frac{1}{2})$

$Z_1 = \sum_{n=0}^{\infty} e^{-\beta\cdot\hbar\omega(n+\frac{1}{2})} = \frac{e^{-\frac{1}{2}\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}}$

$U = -3N\frac{\partial}{\partial\beta}(\ln Z_1)$

$= \frac{3}{2}N\hbar\omega + 3N\cdot\frac{e^{-\beta\hbar\omega}\cdot(\hbar\omega)}{1-e^{-\beta\hbar\omega}}$

$= 3N\hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega}-1}\right)$

$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \left(\frac{\partial U}{\partial\beta}\right)\cdot\left(\frac{\partial\beta}{\partial T}\right)$

$= \frac{1}{kT^2}\,3N\hbar\omega\cdot\frac{e^{\beta\hbar\omega}\cdot\hbar\omega}{(e^{\beta\hbar\omega}-1)^2}$

$= 3Nk\left(\frac{\theta_E}{T}\right)^2\cdot\frac{e^{\theta_E/T}}{(e^{\frac{\theta_E}{T}}-1)^2}$　　$\hbar\omega = k\theta_E$

高温：$T \gg \theta_E$ 时　$C_V \approx 3Nk$

解释：此时能级间距远小于 kT，可忽略能量量子化效应.

低温：$T \ll \theta_E$ 时　$C_V \approx 3Nk\cdot\left(\frac{\theta_E}{T}\right)^2\cdot e^{-\frac{\theta_E}{T}}$

$\lim_{T\to 0} C_V = 0$

解释：此时能级间距远大于 kT，振子冻结在基态，对热容无贡献

§ 7.8 顺磁性固体

配分函数: $Z_1 = e^{\beta\mu B} + e^{-\beta\mu B}$

磁化强度: $m = -\frac{1}{\mu_0 V}\frac{dW}{dH} = -\frac{1}{\mu_0 V}\left(-\frac{N}{\beta}\frac{\partial}{\partial B}\ln Z_1\right)$

$$= \frac{n}{\beta}\frac{\partial}{\partial B}(\ln Z_1)$$

$$= \frac{n}{\beta}\cdot\frac{\mu\beta\left(e^{\beta\mu B}-e^{-\beta\mu B}\right)}{e^{\beta\mu B}+e^{-\beta\mu B}} = \mu n\tanh\left(\frac{\mu B}{kT}\right)$$

弱场/高温极限 $\frac{\mu B}{kT} \ll 1$

$$m \approx \mu n\cdot\frac{\mu B}{kT} = \frac{\mu^2 n B}{kT}$$

$= \chi B$，其中磁化率 $\chi = \frac{\mu^2\mu_0 n}{kT}$ 线性极化.

强场/低温极限 $\frac{\mu B}{kT} \gg 1$

$$m \approx \mu n$$

所有自旋磁矩都沿外场方向，磁化达到饱和

内能密度 $u = -n\frac{\partial}{\partial\beta}(\ln Z_1) = -n\mu B\tanh\left(\frac{\mu B}{kT}\right)$

$$= -mB$$

熵: $S = Nk\left(\ln Z_1 - \beta\frac{\partial(\ln Z_1)}{\partial\beta}\right)$

$$= Nk\left[\ln 2 + \ln\cosh\frac{\mu B}{kT} - \frac{\mu B}{kT}\tanh\frac{\mu B}{kT}\right]$$

弱场/高温极限 $\frac{\mu B}{kT} \ll 1$, $\ln\left(\cosh\left(\frac{\mu B}{kT}\right)\right) \approx \left(\frac{\mu B}{kT}\right)^2$

$$S \approx Nk\left[\ln 2 + \left(\frac{\mu B}{kT}\right)^2 - \left(\frac{\mu B}{kT}\right)^2\right]$$

$= k\ln 2^N$ $\Omega = 2^N$ 每个磁矩有两种可能状态

强场/低温极限 $\frac{\mu B}{kT} \gg 1$

$$S \approx Nk\left(\ln 2 + \frac{\mu B}{kT} - \ln 2 - \frac{\mu B}{kT}\right)$$

$$= \underline{0} \qquad \Omega = 1 \qquad \text{每个磁矩都沿外场方向.}$$

§7.9 负温度状态

习题

7.1

$$\varepsilon = \frac{2\pi^2\hbar^2}{m} \cdot (L^3)^{-2/3} (n_x^2 + n_y^2 + n_z^2)$$

$$\frac{\partial \varepsilon}{\partial V} = \frac{\partial \varepsilon}{\partial L^3}$$

$$= -\frac{2}{3} \frac{2\pi^2\hbar^2}{m} \cdot (L^3)^{-5/3} (n_x^2 + n_y^2 + n_z^2) .$$

$$= -\frac{2}{3} \cdot \frac{2\pi^2\hbar^2}{mL^5} (n_x^2 + n_y^2 + n_z^2)$$

$$= -\frac{2}{3} \cdot \frac{2\pi^2\hbar^2}{mL^5} (n_x^2 + n_y^2 + n_z^2)$$

$$U = \sum_l a_l \varepsilon_l$$

$$= \frac{2\pi^2\hbar^2}{mL^2} \sum_{n_x, n_y, n_z} (n_x^2 + n_y^2 + n_z^2)$$

$$p = -\sum_l a_l \frac{\partial \varepsilon_l}{\partial V}$$

$$= \frac{2}{3} \cdot \frac{1}{L^3} \cdot \frac{2\pi^2\hbar^2}{mL^2} \sum_{n_x, n_y, n_z} (n_x^2 + n_y^2 + n_z^2)$$

$$= \frac{2}{3} \frac{U}{V}$$

7.2

$$\varepsilon = 2\pi c\hbar (L^3)^{-1/3} (n_x^2 + n_y^2 + n_z^2)^{1/2}$$

$$\frac{\partial \varepsilon}{\partial V} = \frac{\partial \varepsilon}{\partial L^3} = -\frac{1}{3} \frac{2\pi c\hbar}{L^4} (n_x^2 + n_y^2 + n_z^2)^{1/2}$$

$$U = \sum_l a_l \varepsilon_l$$

$$= \frac{2\pi c\hbar}{L} \sum (n_x^2 + n_y^2 + n_z^2)^{1/2}$$

$$p = -\sum_l a_l \frac{\partial \varepsilon_l}{\partial V}$$

$$= \frac{1}{3} \cdot \frac{1}{L^3} \cdot \frac{2\pi c\hbar}{L} \sum (n_x^2 + n_y^2 + n_z^2)^{1/2}$$

$$= \frac{1}{3}\frac{U}{V}$$

7.3

$$Z_1 = \sum \omega_l e^{-\beta\varepsilon_l}$$

$$Z_1^* = \sum \omega_l e^{-\beta\varepsilon_l^*} = \sum \omega_l e^{-\beta(\varepsilon_l+\Delta)}$$

$$= e^{-\beta\Delta} Z$$

$$U = -N \frac{\partial(\ln Z_1)}{\partial\beta}$$

$$U^* = -N \frac{\partial(\ln Z_1^*)}{\partial\beta} = -N \frac{\partial \ln(e^{-\beta\Delta} Z_1)}{\partial\beta}$$

$$= -N \frac{\partial \ln Z_1}{\partial\beta} + N\Delta = U + N\Delta$$

$$Y = -\frac{N}{\beta} \frac{\partial(\ln Z_1)}{\partial y}$$

$$Y^* = -\frac{N}{\beta} \frac{\partial(\ln Z_1^*)}{\partial y} = -\frac{N}{\beta} \frac{\partial \ln(e^{-\beta\Delta} Z_1)}{\partial\beta}$$

$$= -\frac{N}{\beta} \frac{\partial \ln Z_1}{\partial y} + \frac{N}{\beta}\Delta$$

$$S = Nk\left(\ln Z_1 - \beta \frac{\partial(\ln Z_1)}{\partial\beta}\right)$$

$$S^* = Nk\left(\ln Z_1^* - \beta \frac{\partial \ln Z_1^*}{\partial\beta}\right) = Nk\left[\ln(e^{-\beta\Delta} Z_1) - \beta \frac{\partial \ln(e^{-\beta\Delta} Z_1)}{\partial\beta}\right]$$

$$= Nk\left(\ln Z_1 - \beta \frac{\partial \ln Z_1}{\partial\beta} - \beta\Delta + \beta\Delta\right)$$

$$= Nk\left(\ln Z_1 - \beta \frac{\partial \ln Z_1}{\partial\beta}\right) = S$$

7.4.

$$S = k \ln \Omega \qquad \Omega = \frac{N!}{\prod_l a_l!} \prod_l \omega_l^{a_l}$$

$$=$$

$$S = Nk\left(\ln Z_1 - \beta \frac{\partial(\ln Z_1)}{\partial\beta}\right) \qquad Z_1 = \frac{e^{-\beta\varepsilon_s}}{P_s}$$

$$= Nk \; -\beta\varepsilon_s - \ln Z_1 - \beta \cdot$$

7.5.

$$S = k\ln\frac{N!}{(Nx)!\,[N(1-x)]!} = k\{\ln N! - \ln(Nx)! - \ln[N(1-x)]!\}$$

$$= k\{N(\ln N - 1) - Nx[\ln(Nx) - 1] - N(1-x)[\ln N(1-x) - 1]\}$$

$$= k\{N\ln N - Nx(\ln N + \ln x) - N(1-x)[\ln N + \ln(1-x)]\}$$

$$= -kN[x\ln x + (1-x)\ln(1-x)]$$

7.6.

(a) 正常原子分布状态数为 $\Omega_1 = \binom{N}{N-n} = \frac{N!}{(N-n)!\,n!}$

间隙原子分布状态数为 $\Omega_2 = \binom{N}{n} = \frac{N!}{(N-n)!\,n!}$

$$S = k\ln\Omega = k\ln(\Omega_1\cdot\Omega_2)$$

$$= 2k\ln\frac{N!}{(N-n)!\,n!}$$

(b)

$$S \approx 2k\{N(\ln N - 1) - (N-n)[\ln(N-n) - 1] - n(\ln n - 1)\}$$

$$= 2k[N\ln N - (N-n)\ln(N-n) - n\ln n]$$

$$F = nu - TS \qquad (x\ln x)' = 1 + \ln x$$

$$= nu - 2kT[N\ln N - (N-n)\ln(N-n) - n\ln n]$$

$$\frac{\partial F}{\partial n} = u + 2kT[-1 - \ln(N-n) + 1 + \ln n]$$

$$= u + 2kT\ln\frac{n}{N-n} \approx u + 2kT\ln\frac{n}{N}$$

解得 $n = Ne^{-\frac{u}{2kT}}$

7.7.

$$\Omega = \binom{N}{n} = \frac{N!}{(N-n)!\,n!}$$

$$S = k\ln\Omega = k\ln\frac{N!}{(N-n)!\,n!}$$

$$\approx k\{N(\ln N - 1) - (N-n)[\ln(N-n) - 1] - n(\ln n - 1)\}$$

$$= k[N\ln N - (N-n)\ln(N-n) - n\ln n]$$

$$F = nW - TS$$

$$= nW - kT[N\ln N - (N-n)\ln(N-n) - n\ln n]$$

$$\frac{\partial F}{\partial n} = W + kT\ln\left(\frac{n}{N-n}\right) \approx W + kT\ln\left(\frac{n}{N}\right) = 0$$

得 $n = N e^{-\frac{W}{kT}}$

2.8

7.9. 跳过

7.10

$$\bar{\varepsilon} = \int \frac{p^2}{2m} e^{-\alpha - \frac{\beta}{2m}[p_x^2+p_y^2+(p_z-p_0)^2]} \frac{V\,dp_x\,dp_y\,dp_z}{h^3}$$

$$= \frac{Ve^{-\alpha}}{2mh^3} \iint \left(\int_{-\infty}^{+\infty} (p_x^2+p_y^2+p_z^2)\, e^{-\frac{\beta}{2m}p_x^2}\, dp_x\right) e^{-\frac{\beta}{2m}(p_y^2+p_z^2)}\, dp_y\, dp_z$$

$$= \frac{Ve^{-\alpha}}{mh^3} \int\int_{-\infty}^{+\infty} \left[\frac{\sqrt{\pi}}{4(\frac{\beta}{2m})^{3/2}} + (p_y^2+p_z^2)\frac{\sqrt{\pi}}{2(\frac{\beta}{2m})^{1/2}}\right] e^{-\frac{\beta}{2m}p_y^2}\, dp_y\; e^{-\frac{\beta}{2m}(p_z-p_0)^2}\, dp_z$$

$$= \frac{2Ve^{-\alpha}}{mh^3}\int_{-\infty}^{+\infty}\left[\frac{\pi}{8(\frac{\beta}{2m})^2} + \frac{\pi}{8(\frac{\beta}{2m})^2} + p_z^2 \cdot \frac{\pi}{4(\frac{\beta}{2m})}\right] e^{-\frac{\beta}{2m}(p_z-p_0)^2}\, dp_z$$

$u_z = p_z - p_0$

$$= \frac{2Ve^{-\alpha}}{mh^3}\int_{-\infty}^{+\infty}\left[\frac{\pi}{8(\frac{\beta}{2m})^2} + \frac{\pi}{8(\frac{\beta}{2m})^2} + (u+p_0)^2 \cdot \frac{\pi}{4(\frac{\beta}{2m})}\right] e^{-\frac{\beta}{2m}u^2}\, du$$

$$= \frac{4Ve^{-\alpha}}{mh^3}\int_0^{+\infty} \left(\frac{\pi m^2}{\beta^2} + \frac{\pi m p_0}{\beta} u + \frac{\pi m}{2\beta}u^2\right) e^{-\frac{\beta}{2m}u^2}\, du$$

$$= \frac{4Ve^{-\alpha}}{mh^3}\left[\frac{\pi m^2}{\beta^2}\cdot\frac{\sqrt{\pi}}{2(\frac{\beta}{2m})^{1/2}} + \frac{\pi m p_0}{\beta}\cdot\frac{1}{2(\frac{\beta}{2m})} + \frac{\pi m}{2\beta}\cdot\frac{\sqrt{\pi}}{4(\frac{\beta}{2m})^{3/2}}\right] e^{-\frac{\beta}{2m}u^2}\, du$$

$$= \frac{4\pi^{3/2}m^{3/2}Ve^{-\alpha}}{h^3\beta^{5/2}}\left(\frac{\sqrt{2}}{2} + \sqrt{\frac{\beta}{2\pi m}}\,p_0 + \frac{\sqrt{2}}{4}\right)$$

$$= \frac{4\pi^{3/2}m^{3/2}Ve^{-\alpha}}{h^3\beta^{5/2}}\left(\frac{3}{4}\sqrt{2} + \sqrt{\frac{\beta}{2\pi m}}\,p_0\right)$$

$e^{-\alpha}$ 可以通过归一化条件求得：

$$\frac{e^{-\alpha}V}{h^3}\int_{-\infty}^{+\infty} e^{-\frac{\beta}{2m}p_x^2}\,dp_x \cdot \int_{-\infty}^{+\infty} e^{-\frac{\beta}{2m}p_y^2}\,dp_y \cdot \int_{-\infty}^{+\infty}e^{-\frac{\beta}{2m}(p_z-p_0)^2}\,dp_z$$

$$= \frac{8e^{-\alpha}V}{h^3}\left[\frac{\sqrt{\pi}}{2(\frac{\beta}{2m})^{1/2}}\right]^3 = \frac{2\sqrt{2}\, e^{-\alpha}V\pi^{3/2}m^{3/2}}{h^3\beta^{3/2}} = 1$$

$$\Rightarrow e^{-\alpha} = \frac{\sqrt{2}\, h^3\beta^{3/2}}{4V\pi^{3/2}m^{3/2}}$$

代回：$\bar{\varepsilon} = \frac{\sqrt{2}}{\beta}\left(\frac{3}{4}\sqrt{2} + \sqrt{\frac{\beta}{2\pi m}}\,p_0\right)$

$$= \underbrace{\frac{3}{2} kT}_{\text{热运动}} + \underbrace{\sqrt{\frac{kT}{3m}}\, p_0}_{\text{平动附加}}$$

7.11

$$f(v_x, v_y)\, dv_x\, dv_y = \frac{m}{2\pi kT} e^{-\frac{1}{2}m(v_x^2+v_y^2)} \cdot dv_x\, dv_y$$

$$\hookrightarrow f(v)\, dv = \frac{m}{kT} \cdot e^{-\frac{1}{2}mv^2}\, v\, dv$$

$\bar{v} =$

$v_p =$

$v_s =$

7.12

2.13